

Pathological Changes in the Intestine of the Marine Fish *Polydactylus sexfilis* (Val. 1831), due to Heavy Infestation of the Acanthocephalan Parasite, *Raorhynchus polynemi* from Eastern Coast of Bay of Bengal

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ABSTRACT The study describes the histopathological manifestations due to acanthocephalan, *Raorhynchus polynemi* in the intestinal of the marine threadfin fish, *Polydactylus sexfilis*, a commercially valued food fish along the east coast of India. The fish collected from Visakhapatnam coast, Bay of Bengal harbored the acanthocephalan parasite, *R. polynemi* with up to 95.6% prevalence. Heavy infestations with the parasites were observed in the anterior two-thirds of the intestine, almost clogging the intestinal lumen. The morphological and diagnostic characteristics of *R. polynemi* were studied by means of light microscopy while their pathological effects on the host intestinal tissues were assessed through transmission electron microscopy. The parasites penetrated through the intestinal layers with their proboscis and neck, while their trunks hanged in the intestinal lumen absorbing the nutrients from the host intestine. At the point of parasite attachment, some of the layers were indistinct or absent, villar epithelium and parts of the enclosed lamina propria were ruptured at contact point with the worm's proboscis hooks. In heavy infections, at the site of perforation, all the layers of the intestine were completely injured as the parasite caused a lot of damage. The electron microscopy observations revealed the accumulation of heterophilic granulocytes, few lymphocytes, macrophages, and mucus droplets near the presomal worm surface and the increase in number of goblet cells, polymorphic nucleocytes, wandering leukocytes, lymphocytes, and granulocytes around the damaged area which is an indication of strong host response. In spite of substantial damage caused to the architecture of the intestinal tissues, there were no noticeable ailing effects on the general health status of the fish and they survived. The parasite was also recovered from the other polynemid fishes, *Polydactylus sextarius*, *P. polynemi*, and *Eleutheronema tetradactylum*.

KEYWORDS Acanthocephala, Histopathology, *Polydactylus sexfilis*, *Raorhynchus polynemi*, Visakhapatnam coast

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INTRODUCTION

Acanthocephalans are a group of endoparasitic helminths with a heavily armed protractible proboscis which can cause mechanical damage to the host, though not feeding on tissues because of lack of digestive system (Olurin and Somorin, 2006). Disease manifestation due to parasites has

become a major restraint in marine Aquaculture (Bondad-Reantaso *et al.*, 2005; Sanil *et al.*, 2010). Parasites on the host cause series of interactions, which ultimately reduce the absorption and interrupt other metabolic processes. Gut dwelling endoparasitic helminths often induce changes in the morphology of the host intestinal tissue and can cause an internal stress that led to modification of the structure and

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function of local tissues (Castro, 1992). The changes brought by the parasite in the host may be due to the mechanical damage or due to the release of toxins by the parasite. Further, the histological changes induced by enteric parasites are also associated with biochemical and physiological changes (Fairweather, 1997). Thus, a comparative study of histology is always useful to distinguish abnormal cells from normal ones, which helps in diagnosis of many diseases (Majumdar, 1980 and Chubb, 1982). Histopathology in fishes resulting from acanthocephalan infections has been reported by a number of earlier researchers who have reported various aspects of impacts due to such parasitization of a number of fishes Ekbaum, 1938; Venard and Warfel, 1953; Esch and Huffines, 1973; Hine and Kennedy, 1974a; 1974b; Chaicharan and Bullock, 1967; Bullock, 1963; Amin and Heckmann, 1992; De Buron and Nickol, 1994; Yildiz *et al.*, 2004; Sanil *et al.*, 2010; Nabi *et al.*, 2015; Sakthivel *et al.*, 2016; Palaq *et al.*, 2017; Sheema, 2018; Bazh and Hamouda, 2019; Dezfali *et al.*, 2021; and Amin and Heckmann, 1992). Necrosis, infiltration of macrophages, and collagenous fiber productions are related by the authors at the point of proboscis attachment. The histopathological investigation by de Lima *et al.*, (1990) on *Piaractus mesopotamicus* parasitized with *Metechinorhynchus jucundus* included an inflammatory reaction with mononuclear cells and hemorrhages the intestinal epithelium. There are no reports of the effects of acanthocephalan infections in members of the family Polynemidae. *Raorhynchus polynemi* Tripathi, 1959 is a common acanthocephalan parasite of Polynemidae fish. The infection level of this parasite in members of the family: Polynemidae is extremely high. *R. polynemi* localized mainly in the anterior part of the intestine of *Polydactylus sexfilis*, reaching a prevalence of up to 100% of the population. In the present study, the pathological effects of *R. polynemi* on the intestine of *P. sexfilis* have been evaluated histologically.

MATERIAL AND METHODS

Study Site

Marine threadfin fish of the family Polynemidae, *P. sexfilis* (Valenciennes) were collected from boat seine and trawlers of fishing harbor, fish landing centers and fish markets in and around Visakhapatnam coast (17.67°N and 83.32°E), Bay of Bengal between July 2005 and June 2007 for a period of 2 years. The fishes were dead at the time of the collection. Morphometric and meristic information of the hosts were recorded on specially prepared data sheets, along with their spectrum of metazoan parasites. Before dissecting the fish, its length, weight and sex were noted. The different parts of the gastrointestinal tract of the host fish were screened for endoparasites. Acanthocephalans were found to be heavily infesting the intestine of the host fish, *P. sexfilis*. The ethics and consent concerning the use of fish for this research were deemed unnecessary according to the Animal ethics committee Control and Supervision of Experiments on Animals as the fish was dead at the time of collection, edible and was easily available to carry out research.

Whole Mounts

For preparing whole mounts, specimen was pressed slightly between glass slides, fixed in alcohol-formaldehyde-acetic

acid overnight, stained in Alum carmine stain and washed thoroughly and dehydrated by using a graded sequence of ethanol's (70%, 80%, 95%, 100%) cleared in xylene and finally mounted in Canada balsam or DPX mount on a glass slide with coverslip and proper care is taken to avoid formation of any air bubbles during mounting. The preparation of permanent slides were followed vide Hiware *et al.* (2003). The stained parasites were observed and identified under the Nikon Alphashot microscope (YS 2) and their micro photographs were captured and line diagrams were drawn with the aid of attached drawing tube.

Histopathology

For histopathology work, both the infected and uninfected intestinal tissues of the host fish were fixed in standard fixatives such as Bouin's fluid and Susa solution for 24 h for the histopathological studies. Post-fixation was followed by following the routine standard methods, clearing in xylene and embedding in paraffin wax (58–62°C). The blocked tissues were sectioned at 4–5 microns in the Rotary microtome (RM2235) and stained in Eosin Hematoxylin double staining method and finally mounted using Canada balsam or DPX mountant (Pearse, 1968, Bancraft, 1975). Sections of the tissues on the slides were examined under the Nikon Alphashot microscope and photomicrographs were taken.

Transmission electron microscopy

For the electron microscopic studies, the samples were fixed in 2.5% Glutaraldehyde in 0.05M Phosphate buffer (pH 7.2) for 24 h at 4°C and post fixed in 0.5% aqueous osmium tetroxide in the same buffer for 2 h. After the post fixation samples were dehydrated in a series of graded alcohol, infiltrated, and embedded in Spurr's resin (Spurr, 1969). Sections were cut with a glass knife on a Leica Ultra cut UCT-GA-D/E-1/00 ultra-microtome, semi thin section of 200–300 nm thick were stained with toluidine blue and ultra-thin sections (50–70 nm thickness) were mounted on grids. Then the ultra-thin sections were stained with saturated aqueous Uranyl acetate and counter stained with 4% lead citrate. These stained sections were observed at various magnifications under transmission electron microscope. Electron microscopic methods of Bozzola and Russell (1999) have been applied for the present study. Electron microscopic methods were borrowed from laboratory of Acharya N.G. Ranga Agricultural University, Hyderabad, where the entire process including photography was taken.

RESULTS

The small intestine of the adult specimens of *P. sexfilis* was found to be heavily parasitized by *R. polynemi* (Acanthocephala: Neo echino rhynchidae) reaching a prevalence rate of 95.6% (110/115). A total of 725 *R. polynemi* (range=1-45) with a mean intensity of 6.59 and mean abundance of 6.3 were recovered from these infected hosts (Figs. 1-3).

Taxonomic Status

Parasite: *R. polynemi* Tripathi, 1959 belongs to family Neo echino rhynchidae.



Fig. 1. Intestine of fish showing *acanthocephalan* worms

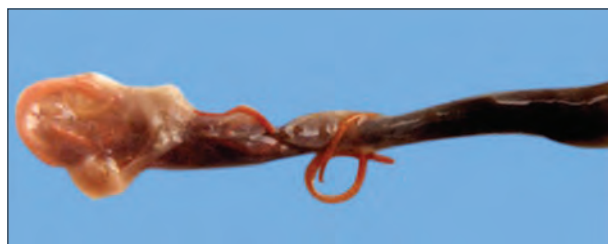


Fig. 2. *Raorhynchus polynemi* in situ as viewed from luminal surface



Fig. 3. *Raorhynchus polynemi* collected in a Petri dish

Type host: Marine threadfin, *P. sexfilis* belonging to Polynemidae family.

Site of infection: Intestine.

Host Locality: Visakhapatnam coast (17.67°N and 83.32°E), Bay of Bengal located in Andhra Pradesh, India.

Prevalence of infection: 95.6% (110 out of 115 examined fish)

Material deposition: Voucher specimens deposited in the Department of Zoology, Andhra University, Visakhapatnam, Andhra Pradesh, India.

Microscopic examination (Figs. 4-6c):

***R. polynemi* Tripathi, 1956** (Synonym: *R. polynemi* Gupta and Lata, 1967).

Male (based on 20 measured specimens): Body uniformly thick measuring 5.0–5.05 × 0.47–0.49. Proboscis cylindrical, elongate and measures 0.58–0.60 × 0.08–0.11, with 11–12 longitudinal rows of 22–24 hooks each, similar in shape but different in size. Hooks on dorsal side smaller, thicker, and more curved than the slenderer ventral hooks. Both dorsal and ventral hooks are smaller posteriorly. The hooks of the last row are abruptly larger than those of penultimate row and arcuate. Neck small and without spines. Trunk spines present anteriorly,

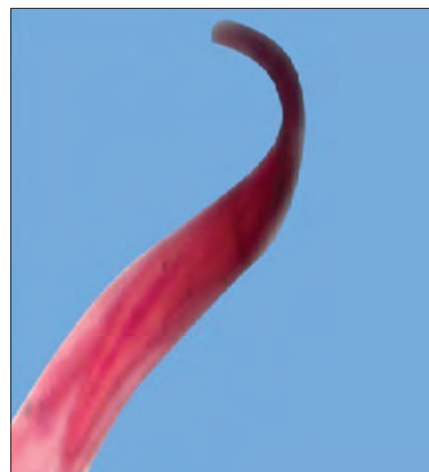


Fig. 4. Microphotograph of anterior region of *Raorhynchus polynemi*, ×40



Fig. 5. Microphotograph of middle region of *Raorhynchus polynemi*, ×40

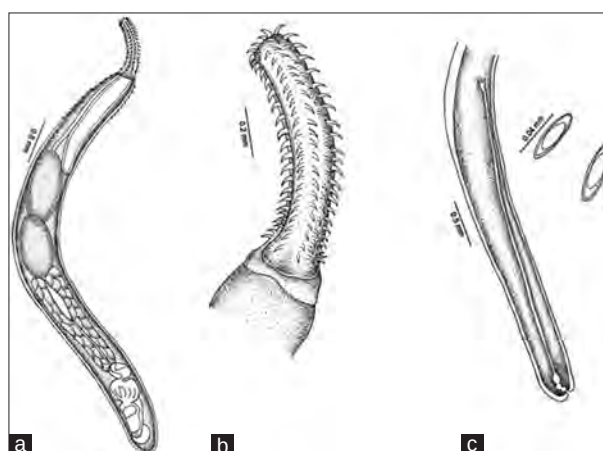


Fig. 6. *Camera lucida* drawing of *Raorhynchus polynemi*, ×40 (a) *C. lucida* drawing of Proboscis-enlarged, ×100 (b) *C. lucida* drawing of Posterior region, ×40 (c) *C. lucida* drawing of eggs enlarged, ×600

with 8 rows dorsally and 12 rows ventrally. Proboscis receptacle claviform, double layered measuring 1.24–1.26 × 0.21–0.24. Lemnisci equal and slightly smaller than proboscis

receptacle measuring $1.32\text{--}1.34 \times 0.05\text{--}0.08$. Testes two, oval, tandem and reaching half-way upto the proboscis receptacle. Anterior testis longer measuring $0.79\text{--}0.82 \times 0.34\text{--}0.37$ and posterior smaller than anterior, $0.71\text{--}0.74 \times 0.37\text{--}0.39$ in size. Cement glands two, $1.26\text{--}1.29$ in length, tubular, with their thick ducts opening near the penis. Seminal vesicle small, situated near bursa and circular when everted. Bursa elongated and muscular, opening terminally.

Female (based on 20 measured specimens): Body slender, cylindrical, much longer than males measuring $0.63\text{--}0.68 \times 0.39\text{--}0.42$, slightly tapering posteriorly. Proboscis cylindrical. Shape, size and arrangement of proboscis hooks and trunk spines same as in males. Proboscis receptacle double layered. Genital tract long, tubular occupying almost the posterior half of the trunk. Uterine bell funnel shaped. Uterus long, tubular, convoluted and measures $1.89\text{--}1.92$. Vagina with sphincter muscles and opening laterally. Eggs fusiform, with distinct polar bulbs measuring $0.20\text{--}0.23 \times 0.005\text{--}0.008$ with thicker middle shell.

Histopathology (Figs. 7-12)

General pathology: *R. polynemi* is an acanthocephalan with well-developed proboscis with 16–18 rows of pointed hooks. These worms embed themselves into the intestine with the help of spiny proboscis (Fig. 7). In infected fish, the parasites are in the anterior two thirds of the intestine. The infections by this species cause damage to the host by penetration of proboscis, and neck through the intestinal layers, while the trunk of the parasite remained in the intestinal lumen (Fig. 8). The parasites lack gut like cestodes and the body in the lumen absorbs the digested contents, thereby depleting nutrients of the host. Proboscis penetrates completely through the wall of the intestine, but outwardly no nodule or capsule formation is seen (Fig. 9). At the point of attachment of the parasite, some of the layers are indistinct or absent. The unicellular serosal layer and muscularis layer are intact; the third layer stratum compactum acts as a barrier to penetration (Fig. 10). Widespread destruction of the intestinal wall at the site of attachment and adjacent villi are noticed in the infected intestine. In some cases, the proboscis is seen between a few villi, consequently the villar epithelium and parts of the enclosed lamina propria are ruptured at contact point with the worm's proboscis hooks. At some sites, the mucosa is completely destroyed but the lamina propria is only partially damaged. In heavy infection, at the site of perforation, mucosa, lamina propria, stratum compactum, muscle layer and serosa have been completely disrupted (Fig. 11) and the parasite caused lots of damage. Epithelium fragmented into a mass of compressed, markedly degenerated cells.

Transmission Electron Microscopy (Figs. 12 and 13)

However, the electron microscopic observations revealed that the inflammatory tissue of the host contained many heterophilic granulocytes and also few lymphocytes and macrophages and also mucus like droplets which accumulate near the presomal worm surface. The goblet cells in the mucosa region were more prominent and their number increases (Fig. 11) when compared to the undamaged villus. In the affected villi the remaining lamina propria becomes

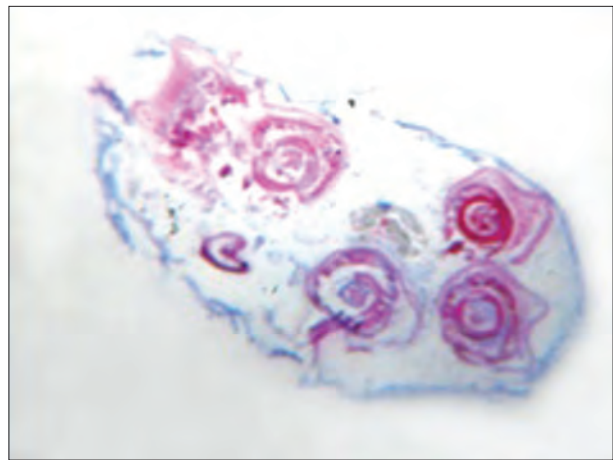


Fig. 7. T.S. of infected intestine showing parasites, $\times 40$

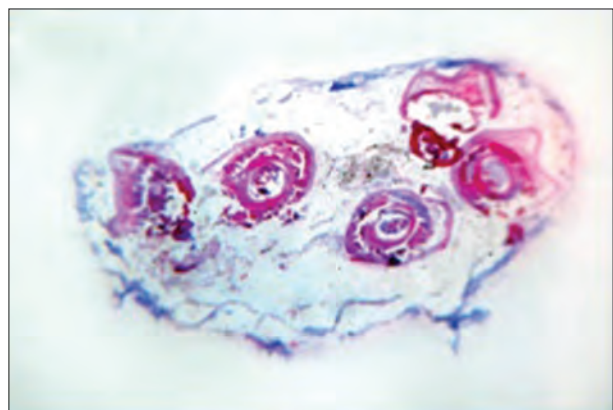


Fig. 8. T.S. of infected intestine showing parasites and ruptured intestinal layers, $\times 40$

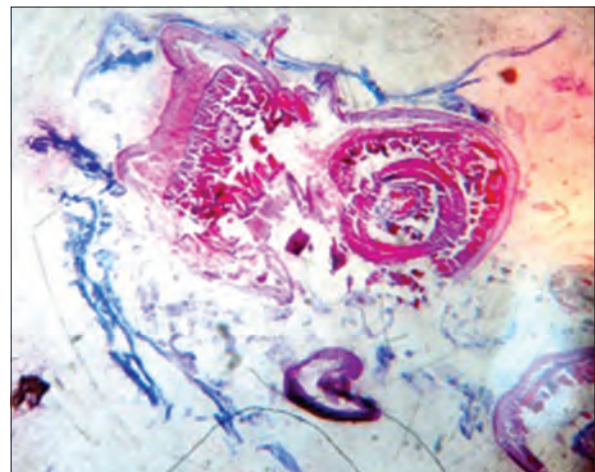


Fig. 9. T.S. of infected intestine showing proboscis attached in the lamina propria, $\times 100$

hemorrhagic and defense cells and other necrotic cells are seen discharged from the lesions. Erythrocytes and leukocytes are present in enormous numbers at the site of damage. The goblet cells are undifferentiated because all tissue at the site of penetration had lost differentiation into layers (Fig. 12). Polymorphic nucleocytes, each with a five lobed, horse shoe shaped nucleus, small wandering leukocytes or lymphocytes

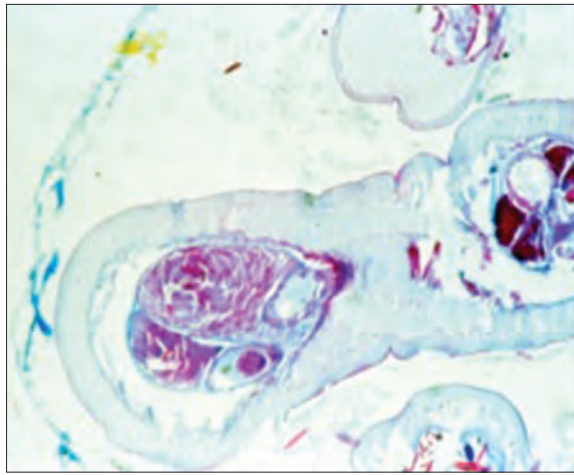


Fig. 10. T.S. of infected intestine containing worms, $\times 100$

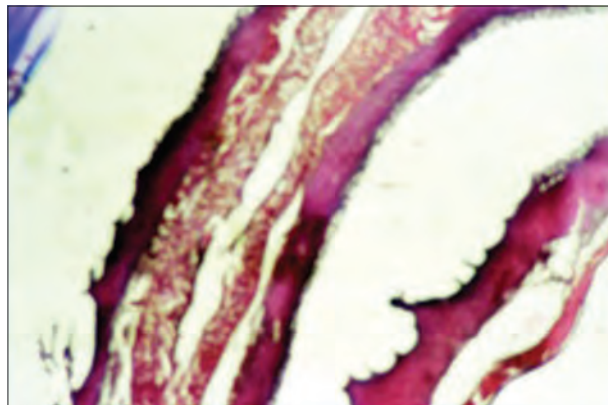


Fig. 11. T.S. of infected intestine showing proboscis damaging mucosal crypts, $\times 100$

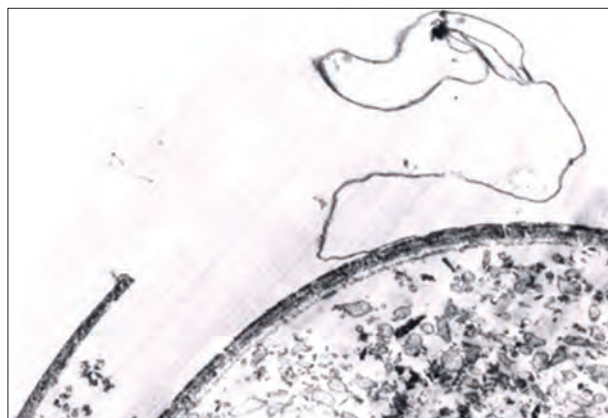


Fig. 12. Electron microphotograph of infected intestine showing complete desquamation, $\times 7720$

and other granulocytes are observed in greater numbers around the area of damage (Fig. 13). Histopathological studies revealed a severe pathological alterations and mechanical damage caused by the parasites have absolutely destroyed the architecture of the intestinal tissues. In spite of the pathological changes caused by the worms to the intestinal tissues, there were no visible ill effects on the general health status of the host fish.

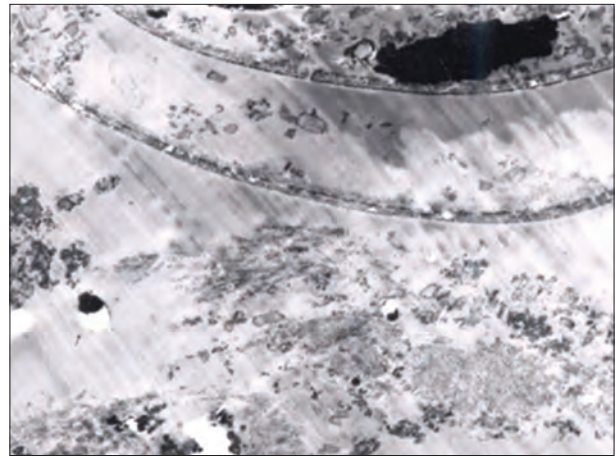


Fig. 13. Electron microphotograph of intestinal epithelium of the host showing hyperplasia and hypertrophy of goblet cells, $\times 5790$

DISCUSSION

This is the first report of *R. polynemi* heavily infecting the *P. sexfilis*, causing severe pathological disorders in the infected host. Although there are conflicting views regarding the extent of damage caused by acanthocephalans, it is agreed that these worms are harmful to the hosts because of their strong-armed proboscis. The degree of pathology and intensity of host reaction is dependent upon the species of parasite and upon the site of localization as reported earlier workers Venard and Warfel, 1953; Esch and Huffines, 1973; Hine and Kennedy, 1974a, 1974b and Feist and Longshaw, 2008). Even the nature and depth of various tissue layers, length of the neck and proboscis, presence or absence of proboscis bulb and arrangement of spines can affect the pathological outcome. Multiple wound formation in trout intestine parasitized by *Acanthocephalus jacksoni* has been reported by Bullock (1963) and Chaicharan and Bullock (1967) on catostomid fishes parasitized by *Neocalanus cristatus*, *N. prolixoides* and male *Octospinifer macilentus*. The natural infection of *P. sexfilis* with *R. polynemi* resulted in definite histopathology; however, the wound formation was not noticed. The tunnels filled with cores of debris and blood noted by Venard and Warfel (1953) in the intestinal wall of *Huro salmoides* parasitized by *Leptorhynchoides thecatus* was also not observed during the present study. There are many histopathological reports regarding the perforation of the intestinal wall by different genera of acanthocephalan parasites in various hosts (Wurmbach, 1937; Takos and Thomas, 1958; Prakash and Adams, 1960; Bullock, 1963; Raina and Koul, 1984; Wali *et al.*, 2006; Castro *et al.*, 2006; Jithendran and Kannapan, 2010; de Matos *et al.*, 2017; Verma and Saxena, 2018; and Castro *et al.*, 2020). In the present report, perforation is observed in the intestine of the host. In most instances, all layers of the intestinal wall are penetrated by the proboscis and neck of *R. polynemi* while the trunk remained in the lumen. When compared to the normal, the infected anterior region of the intestine appeared puffy and sagging which can be accredited to the weakened circular and longitudinal muscle layers of the intestine capitulating to the compression from heavy load of parasites inside the

lumen. Intense host responses were observed in most areas of contact between the parasite and the host. In heavy infections, the intestine appears to be crowded with the worms, almost blocking the intestinal lumen, thereby adversely affecting the movement of the digested materials within the intestine and the absorption of nutrients. A similar host response to an immobile parasite was reported by Hine and Kennedy (1974a, 1974b) for *Pomphorynchus laevis* in salmonid and cyprinid fishes. Increase in the number of goblet cells reported by Bullock (1963) in salmonid fishes infected with *A. jacksoni* was also observed in the present study. Esch and Huffines (1973) studied the tissue reaction of *Micropterus dolomieu* infected with *L. thecatus*. A brief description of the histopathological aspects caused by *M. jacundus* in *P. mesopotamicus* is given by de Lima *et al.* (1990). These authors observed inflammatory foci of mononuclear cells and hemorrhages in the intestine. Similarly, with the deep penetration of the parasite proboscis, fibrosis, eosinophilic granulocytosis, complete desquamation of the epithelium of the intestinal mucosa, severe hyperplasia and hypertrophy of the goblet cells were observed which is indicative of intense host response. Intestinal obstruction, compression of villi, hemorrhage, presence of macrophages and fibroblasts are the tissue changes reported by Amin and Heckmann (1992). The present study also confirms these results, in which the proboscis attachment provoked the intense host tissue response. According to Taraschewski (2000), the pathogenicity of acanthocephalans was caused by two factors, that is, depth of the parasite penetration into the host tissue and the density of parasites. Although the pathological effects are confined around the individual worms, in the present case the total destruction of the tissue architecture happened due to the unusual heavy parasitic load. The present study is in total confirmation with the views of Hine and Kennedy (1974a, 1974b) who observed that fish appeared healthy without any clinical manifestations. Thus, the histopathological observations as a whole, affect the health background of the host fish resulting in the depletion of muscle quality, fish growth, susceptibility for other diseases, etc. Although one cannot suggest remedial programs for a marine fish health, it is only interesting and important observation to record which suggests that there may be depletion of muscle quality in parasite infected fish if not mortality. The damage in intestinal wall results in poor absorption, at the same time the parasite is also competing with the host for the digested nutrients there by affecting the health status of the host.

CONCLUSION

Histopathology of *P. sexfilis* infected with the acanthocephalan, *R. polynemi* causes damage to the host by penetration of proboscis and neck through the intestinal lumen, while the trunk of the parasite remained in the intestinal lumen absorbing the nutrients from the host intestine. The parasites are confined mainly in the anterior two-thirds of the intestine in infected fish. Both the light microscopy and electron microscopy observations reveals complete disruption of all the layers of the intestine and the accumulation of heterophilic granulocytes, goblet cells, polymorphic nucleocytes,

wandering leukocytes, few lymphocytes, macrophages, and mucus droplets around the damaged area which is an indicative of intense host responses. In spite of high level of infection, *P. sexfilis* survives. The damage caused to the host depends on type of spoliation of host tissue, number of parasites and the health status of the host.

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